

Machine Learning Types

Complete Study Note • Video timestamps 00:00 – 22:29+

Big Picture Map

ML types are categorized by one central question: how much labeled data / supervision does the algorithm get during training?

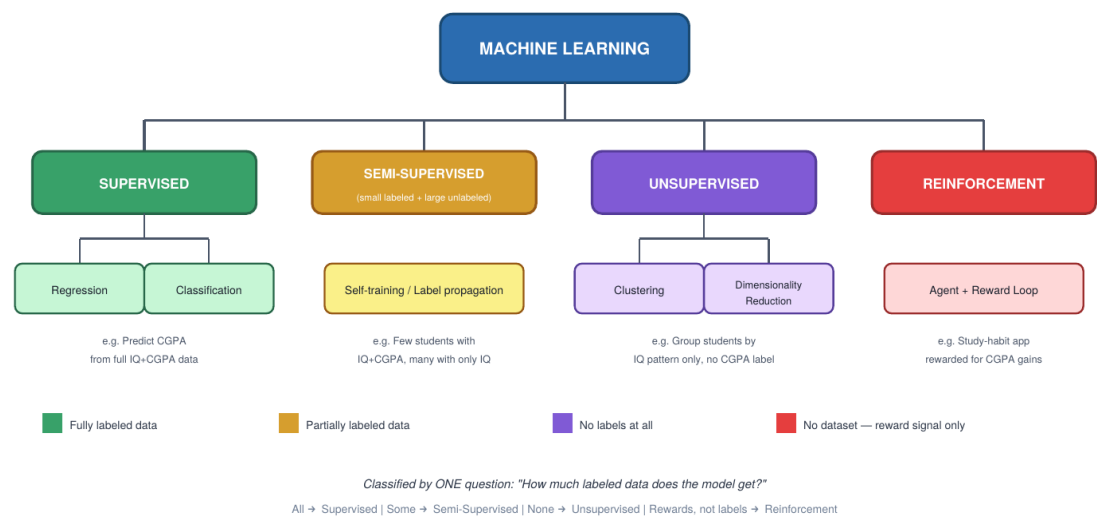


Fig 1. The four branches of Machine Learning, grouped by amount of labeled data available.

Amount of Labeled Data	Type
All data labeled	Supervised
Some labeled, most unlabeled	Semi-Supervised
No data labeled	Unsupervised
No dataset — only rewards	Reinforcement

1. Supervised Learning (06:05)

The model learns from input-output pairs (labeled data). You already know the correct answer for training examples, and the model learns the mapping $X \rightarrow Y$.

Regression

Predicts a continuous numeric value. Example: predicting house price, temperature, or salary.

Classification

Predicts a category or class label. Example: spam vs. not spam, cat vs. dog, disease positive/negative.

Feature	Regression	Classification
Output type	Continuous number	Discrete category
Example	House price (\$245,000)	Email $\rightarrow$ Spam / Not Spam
Common algorithms	Linear Regression, SVR	Logistic Regression, Decision Trees, SVM
Evaluation metric	MSE, RMSE, R^2	Accuracy, Precision, Recall, F1

Mental shortcut: "How much?" $\rightarrow$ Regression. "Which one?" $\rightarrow$ Classification.

2. Semi-Supervised Learning

A middle ground between supervised and unsupervised learning. The model trains on a small amount of labeled data plus a large amount of unlabeled data — used when labeling is expensive or slow, but collecting raw data is easy.

Example: Predicting CGPA from IQ

Getting a student's IQ is quick (a test). Getting their CGPA takes an entire semester to confirm. So the dataset looks like this:

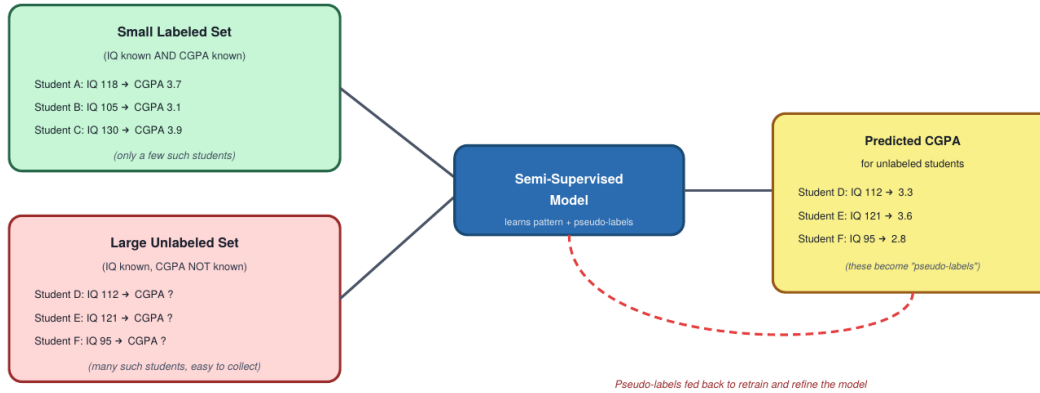
Student	IQ	CGPA	Status
A	118	3.7	Labeled
B	105	3.1	Labeled
C	130	3.9	Labeled
D	112	?	Unlabeled
E	121	?	Unlabeled
F	95	?	Unlabeled

Only 3 students have both IQ and CGPA recorded. Hundreds more have only IQ scores.

How it works

- Train an initial model on the small labeled set (A, B, C).
- Use it to predict ("pseudo-label") CGPA for the unlabeled students (D, E, F).
- Add confident pseudo-labels back into the training set.
- Retrain on the combined real + pseudo-labeled data $\rightarrow$ a better, more generalized model.

Semi-Supervised Learning — Predicting CGPA from IQ



Key idea: A LITTLE labeled data teaches the model the pattern; a LOT of unlabeled data helps it generalize better.

Fig 2. Semi-supervised workflow: small labeled set + large unlabeled set → pseudo-labels → retrained model.

Feature	Supervised	Semi-Supervised	Unsupervised
Labeled data	100%	Small % (5-10%)	0%
Unlabeled data	None	Large % (rest)	100%
IQ/CGPA example	All students have both	Few have CGPA, most only IQ	No CGPA known — group by IQ pattern only
Use case	Spam detection, price prediction	Medical diagnosis, speech recognition	Customer segmentation

Mental shortcut: "I have a little bit of the answer key — can I use it to guess the rest, then learn from my own guesses?" → Semi-Supervised.

3. Unsupervised Learning (11:20)

The model works with unlabeled data — no correct answers given. It finds hidden structure, patterns, or groupings on its own.

Technique	What it does	Example
Clustering	Groups similar data points together	Customer segmentation, grouping news by topic
Dimensionality Reduction	Reduces features while keeping key info	PCA for image compression, 2D/3D visualization

Mental shortcut: No labels, no answer key — the model discovers patterns by itself.

4. Reinforcement Learning (16:45)

An agent interacts with an environment, takes actions, and learns from rewards (positive) and penalties (negative) to maximize long-term reward.

Component	Meaning
Agent	The learner / decision-maker
Environment	The world the agent interacts with
Action	Choice the agent makes
Reward	Feedback signal (positive or negative)
Policy	Strategy the agent learns to maximize reward

Real-world examples:

- Self-driving cars learning to navigate traffic
- Game-playing AI (Chess, Go, video games)
- Robotics (a robot learning to walk)

Mental shortcut: No fixed dataset — the agent learns by trial and error, guided by rewards/penalties over time.

5. Applications & Examples (22:29)

Domain	ML Type Used	Example
Image Recognition	Supervised (Classification)	Identifying objects/faces in photos
Association Rule Learning	Unsupervised	"Bread buyers also buy butter" (market basket analysis)
Deep Reinforcement Learning	Reinforcement + Deep Learning	AlphaGo, advanced robotics, autonomous systems

Key takeaway: ML techniques help uncover hidden patterns in data and support smarter business decisions — from recommending products to automating complex control tasks.

One-Page Summary

Type	Data	Learns By	Output	Examples
Supervised	Fully labeled	Input→Output mapping	Prediction (number/category)	Price prediction, spam detection
Semi-Supervised	Small labeled + large unlabeled	Pseudo-labeling + retraining	Refined prediction	Predicting CGPA from IQ, medical diagnosis
Unsupervised	Unlabeled	Finding structure	Groups/patterns	Customer segmentation, PCA
Reinforcement	None (rewards only)	Trial & error via rewards	Optimal action/policy	Game AI, robotics, self-driving cars

Quick Revision Questions

- What differentiates regression from classification?
- In the IQ/CGPA example, why is this semi-supervised rather than fully supervised?
- What is a "pseudo-label" and how is it generated?
- Give one real-world example each of clustering and dimensionality reduction.
- What are the four core components of a reinforcement learning system?
- Why is association rule learning considered unsupervised?